

KNUST – Level 100 Biochemistry

Exam

Course: FST 154 – Introductory Biochemistry for Food Scientists **Duration:** 2 hours
Answer all questions.

Questions 1 – 50

Q1. Proteins are composed of repeating units of: A) Fatty acids B) Amino acids C) Monosaccharides D) Nucleotides

Q2. Which amino acid has a sulfur-containing side chain? A) Methionine B) Valine C) Glycine D) Alanine

Q3. The peptide bond is formed between: A) Two carboxyl groups B) Two amino groups C) An amino group and a carboxyl group D) Two R-groups

Q4. Which vitamin deficiency causes scurvy? A) Vitamin A B) Vitamin B12 C) Vitamin C D) Vitamin D

Q5. The enzyme model that suggests flexibility of the active site is: A) Lock and Key B) Induced Fit C) Transition State D) Substrate Channeling

Q6. Lipids are generally soluble in: A) Water B) Organic solvents C) Saline solution D) Buffers

Q7. Which of the following is an essential amino acid? A) Glycine B) Alanine C) Lysine D) Serine

Q8. The sweetest naturally occurring sugar is: A) Glucose B) Fructose C) Galactose D) Maltose

Q9. The enzyme urease shows which type of specificity? A) Absolute B) Group C) Linkage D) Stereochemical

Q10. Vitamin B12 deficiency leads to: A) Pellagra B) Beriberi C) Pernicious anemia D) Rickets

Q11. Which of the following is a conjugated protein? A) Albumin B) Hemoglobin C) Keratin D) Collagen

Q12. The structural polysaccharide in plant cell walls is: A) Glycogen B) Amylose C) Cellulose D) Starch

Q13. Which amino acid is achiral? A) Alanine B) Glycine C) Valine D) Leucine

Q14. The enzyme catalase acts specifically on: A) Urea B) Hydrogen peroxide C) Glucose D) Lactose

Q15. Which vitamin is also known as retinol? A) Vitamin A B) Vitamin B1 C) Vitamin C D) Vitamin D

Q16. The main storage polysaccharide in animals is: A) Amylose B) Glycogen C) Cellulose D) Starch

Q17. The peptide hormone insulin contains how many amino acid residues? A) 29 B) 51 C) 17 D) 8

Q18. Which lipid is a precursor for steroid hormones? A) Waxes B) Cholesterol C) Triglycerides D) Phospholipids

Q19. The enzyme hexokinase belongs to which class? A) Oxidoreductases B) Transferases C) Hydrolases D) Lyases

Q20. Which vitamin deficiency leads to rickets? A) Vitamin A B) Vitamin C C) Vitamin D D) Vitamin K

Q21. The amino acid tryptophan is classified as: A) Nonpolar aliphatic B) Aromatic C) Polar uncharged D) Positively charged

Q22. Which carbohydrate is a ketohexose? A) Glucose B) Galactose C) Fructose D) Ribose

Q23. The enzyme-substrate complex is represented as: A) $S + E \rightarrow P$ B) $S + E \rightarrow ES \rightarrow EP \rightarrow P + E$ C) $P + E \rightarrow S$ D) $ES \rightarrow S + P$

Q24. Vitamin K is essential for: A) Vision B) Collagen synthesis C) Blood clotting D) Calcium absorption

Q25. Which amino acid is imino in nature? A) Proline B) Glycine C) Alanine D) Serine

Q26. The bond that links monosaccharides in disaccharides is: A) Peptide bond B) Glycosidic bond C) Hydrogen bond D) Ionic bond

Q27. Which enzyme catalyzes the breakdown of starch? A) Lipase B) Amylase C) Protease D) Maltase

Q28. Vitamin B1 deficiency causes: A) Pellagra B) Beriberi C) Scurvy D) Rickets

Q29. The amino acid histidine is classified as: A) Nonpolar B) Aromatic C) Positively charged D) Negatively charged

Q30. Which lipid is the major component of cell membranes? A) Triglycerides B) Phospholipids C) Waxes D) Sterols

Q31. The enzyme class that catalyzes hydrolytic cleavage is: A) Oxidoreductases B) Transferases C) Hydrolases D) Lyases

Q32. Which vitamin acts as an antioxidant? A) Vitamin A B) Vitamin E C) Vitamin D D) Vitamin K

Q33. The sugar present in RNA is: A) Ribose B) Deoxyribose C) Glucose D) Fructose

Q34. The amino acid cysteine can form: A) Hydrogen bonds B) Disulfide bonds C) Ionic bonds D) Peptide bonds

Q35. Vitamin B3 deficiency leads to: A) Pellagra B) Beriberi C) Scurvy D) Rickets

Q36. The enzyme specificity where the enzyme acts on a particular bond type is: A) Absolute B) Group C) Linkage D) Stereochemical

Q37. Which carbohydrate is a disaccharide? A) Glucose B) Maltose C) Ribose D) Galactose

Q38. The quaternary structure of hemoglobin consists of: A) One polypeptide chain B) Two chains C) Four chains D) Eight chains

Q39. Vitamin D helps in absorption of: A) Iron B) Calcium C) Sodium D) Potassium

Q40. The enzyme that hydrolyzes proteins is: A) Lipase B) Protease C) Amylase D) Maltase

Q41. Which amino acid is aromatic and polar? A) Tyrosine B) Phenylalanine C) Tryptophan D) Valine

Q42. The enzyme class that catalyzes joining of molecules is: A) Oxidoreductases B) Ligases C) Lyases D) Hydrolases

Q43. Which vitamin deficiency leads to night blindness? A) Vitamin A B) Vitamin B1 C) Vitamin C D) Vitamin D

Q44. The sugar found in DNA is: A) Ribose B) Deoxyribose C) Glucose D) Fructose

Q45. The enzyme that adds phosphate to hexoses is: A) Hexokinase B) Amylase C) Protease D) Catalase

Q46. Vitamin B2 is also known as: A) Riboflavin B) Niacin C) Thiamine D) Pyridoxine

Q47. The amino acid glycine is classified as: A) Aromatic B) Nonpolar aliphatic C) Polar uncharged D) Achiral

Q48. The enzyme that catalyzes conversion of pyruvate to acetyl-CoA requires:

A) Vitamin B1 B) Vitamin B12 C) Vitamin C D) Vitamin D

Q49. Which lipid serves as energy storage in animals? A) Phospholipids B)

Triglycerides C) Sterols D) Waxes

Q50. The enzyme class that catalyzes isomerization reactions is: A)

Oxidoreductases B) Transferases C) Isomerases D) Hydrolases

Q51. Which amino acid is positively charged at physiological pH? A) Aspartate B)

Glutamate C) Lysine D) Serine

Q52. The enzyme class that catalyzes oxidation-reduction reactions is: A)

Oxidoreductases B) Transferases C) Hydrolases D) Lyases

Q53. Vitamin B6 is also known as: A) Pyridoxine B) Riboflavin C) Niacin D)

Thiamine

Q54. The carbohydrate storage form in plants is: A) Glycogen B) Amylose C)

Cellulose D) Amylopectin

Q55. Which amino acid is aromatic but nonpolar? A) Tyrosine B) Phenylalanine C)

Tryptophan D) Histidine

Q56. The enzyme sucrase acts on: A) Maltose B) Lactose C) Sucrose D) Cellulose

Q57. Vitamin B2 deficiency leads to: A) Pellagra B) Glossitis C) Rickets D) Scurvy

Q58. The sugar found in ATP is: A) Glucose B) Ribose C) Deoxyribose D)

Fructose

Q59. The enzyme that hydrolyzes lipids is: A) Lipase B) Amylase C) Protease D)

Catalase

Q60. Vitamin A deficiency leads to: A) Night blindness B) Pellagra C) Beriberi D)

Scurvy

Q61. Which amino acid has an imino group? A) Proline B) Glycine C) Alanine D)

Valine

Q62. The enzyme class that catalyzes bond cleavage by elimination is: A)

Oxidoreductases B) Transferases C) Lyases D) Ligases

Q63. Vitamin D deficiency causes: A) Scurvy B) Rickets C) Pellagra D) Beriberi

Q64. The carbohydrate that is a reducing sugar is: A) Sucrose B) Maltose C)

Cellulose D) Amylose

Q65. The amino acid serine is classified as: A) Nonpolar B) Polar uncharged C)

Positively charged D) Negatively charged

Q66. The enzyme lactase acts on: A) Maltose B) Lactose C) Sucrose D) Cellulose

Q67. Vitamin B1 is also known as: A) Riboflavin B) Thiamine C) Niacin D) Pyridoxine

Q68. The sugar found in DNA is: A) Ribose B) Deoxyribose C) Glucose D) Fructose

Q69. The enzyme that hydrolyzes triglycerides is: A) Lipase B) Amylase C) Protease D) Maltase

Q70. Vitamin K is required for: A) Vision B) Blood clotting C) Collagen synthesis D) Calcium absorption

Q71. Which amino acid is aromatic and polar? A) Tyrosine B) Phenylalanine C) Tryptophan D) Valine

Q72. The enzyme class that catalyzes joining of molecules is: A) Oxidoreductases B) Ligases C) Lyases D) Hydrolases

Q73. Vitamin B3 deficiency leads to: A) Pellagra B) Beriberi C) Scurvy D) Rickets

Q74. The carbohydrate storage form in animals is: A) Amylose B) Glycogen C) Cellulose D) Amylopectin

Q75. The amino acid glycine is unique because: A) It is aromatic B) It is achiral C) It is positively charged D) It is negatively charged

Q76. The enzyme catalase acts on: A) Urea B) Hydrogen peroxide C) Glucose D) Lactose

Q77. Vitamin B12 deficiency leads to: A) Pernicious anemia B) Pellagra C) Beriberi D) Rickets

Q78. The sugar found in RNA is: A) Ribose B) Deoxyribose C) Glucose D) Fructose

Q79. The enzyme that hydrolyzes proteins is: A) Lipase B) Protease C) Amylase D) Maltase

Q80. Vitamin E acts as: A) Antioxidant B) Coenzyme C) Hormone D) Structural protein

Q81. Which amino acid is nonpolar aliphatic? A) Valine B) Tyrosine C) Serine D) Aspartate

Q82. The enzyme class that catalyzes hydrolytic cleavage is: A) Oxidoreductases B) Transferases C) Hydrolases D) Lyases

Q83. Vitamin C deficiency leads to: A) Pellagra B) Scurvy C) Beriberi D) Rickets

Q84. The carbohydrate that is a disaccharide is: A) Glucose B) Maltose C) Ribose
D) Galactose

Q85. The amino acid cysteine can form: A) Hydrogen bonds B) Disulfide bonds C)
Ionic bonds D) Peptide bonds

Q86. The enzyme hexokinase adds phosphate to: A) Lipids B) Proteins C) Hexoses
D) Amino acids

Q87. Vitamin B2 is also known as: A) Riboflavin B) Niacin C) Thiamine D)
Pyridoxine

Q88. The sugar found in nucleotides is: A) Ribose B) Deoxyribose C) Glucose D)
Fructose

Q89. The enzyme that hydrolyzes starch is: A) Lipase B) Amylase C) Protease D)
Maltase

Q90. Vitamin D helps in absorption of: A) Iron B) Calcium C) Sodium D)
Potassium

Q91. Which amino acid is aromatic but nonpolar? A) Tyrosine B) Phenylalanine C)
Tryptophan D) Histidine

Q92. The enzyme class that catalyzes isomerization reactions is: A)
Oxidoreductases B) Transferases C) Isomerases D) Hydrolases

Q93. Vitamin A deficiency leads to: A) Night blindness B) Pellagra C) Beriberi D)
Scurvy

Q94. The carbohydrate storage form in plants is: A) Glycogen B) Amylose C)
Cellulose D) Amylopectin

Q95. The amino acid histidine is classified as: A) Nonpolar B) Aromatic C)
Positively charged D) Negatively charged

Q96. The enzyme lactase acts on: A) Maltose B) Lactose C) Sucrose D) Cellulose

Q97. Vitamin B1 deficiency causes: A) Pellagra B) Beriberi C) Scurvy D) Rickets

Q98. The sugar found in DNA is: A) Ribose B) Deoxyribose C) Glucose D)
Fructose

Q99. The enzyme that hydrolyzes triglycerides is: A) Lipase B) Amylase C)
Protease D) Maltase

Q100. Vitamin K is required for: A) Vision B) Blood clotting C) Collagen synthesis
D) Calcium absorption

Q101. Which amino acid is negatively charged at physiological pH? A) Aspartate
B) Lysine C) Histidine D) Valine

Q102. The enzyme class that catalyzes transfer of groups is: A) Oxidoreductases B)
Transferases C) Hydrolases D) Lyases

Q103. Vitamin B6 acts mainly as a coenzyme in: A) Transamination reactions B)
Oxidation reactions C) Hydrolysis reactions D) Isomerization reactions

Q104. The carbohydrate that is a polysaccharide is: A) Glucose B) Maltose C)
Amylose D) Ribose

Q105. Which amino acid is aromatic and essential? A) Tyrosine B) Phenylalanine
C) Glycine D) Serine

Q106. The enzyme maltase acts on: A) Maltose B) Lactose C) Sucrose D) Cellulose

Q107. Vitamin B2 deficiency symptoms include: A) Glossitis and cheilosis B)
Night blindness C) Rickets D) Scurvy

Q108. The sugar found in NAD⁺ is: A) Ribose B) Deoxyribose C) Glucose D)
Fructose

Q109. The enzyme that hydrolyzes nucleic acids is: A) Nuclease B) Lipase C)
Protease D) Amylase

Q110. Vitamin A is important for: A) Vision B) Blood clotting C) Collagen
synthesis D) Calcium absorption

Q111. Which amino acid is sulfur-containing? A) Methionine B) Glycine C)
Alanine D) Valine

Q112. The enzyme class that catalyzes bond formation with ATP hydrolysis is: A)
Oxidoreductases B) Ligases C) Lyases D) Hydrolases

Q113. Vitamin D deficiency leads to: A) Rickets B) Pellagra C) Beriberi D) Scurvy

Q114. The carbohydrate that is a trisaccharide is: A) Raffinose B) Maltose C)
Glucose D) Galactose

Q115. The amino acid threonine is classified as: A) Nonpolar B) Polar uncharged C)
Positively charged D) Negatively charged

Q116. The enzyme cellulase acts on: A) Maltose B) Lactose C) Sucrose D)
Cellulose

Q117. Vitamin B3 is also known as: A) Niacin B) Riboflavin C) Thiamine D)
Pyridoxine

Q118. The sugar found in FAD is: A) Ribose B) Deoxyribose C) Glucose D) Fructose

Q119. The enzyme that hydrolyzes fats is: A) Lipase B) Amylase C) Protease D) Maltase

Q120. Vitamin K deficiency leads to: A) Poor blood clotting B) Night blindness C) Rickets D) Scurvy

Q121. Which amino acid is aromatic and nonpolar? A) Phenylalanine B) Tyrosine C) Tryptophan D) Histidine

Q122. The enzyme class that catalyzes rearrangement of atoms is: A) Oxidoreductases B) Transferases C) Isomerases D) Hydrolases

Q123. Vitamin E deficiency is rare but can cause: A) Fragile red blood cells B) Night blindness C) Pellagra D) Beriberi

Q124. The carbohydrate that is a pentose is: A) Ribose B) Glucose C) Galactose D) Fructose

Q125. The amino acid valine is classified as: A) Nonpolar aliphatic B) Aromatic C) Polar uncharged D) Negatively charged

Q126. The enzyme protease acts on: A) Proteins B) Lipids C) Carbohydrates D) Nucleic acids

Q127. Vitamin B12 is also known as: A) Cobalamin B) Riboflavin C) Niacin D) Thiamine

Q128. The sugar found in RNA nucleotides is: A) Ribose B) Deoxyribose C) Glucose D) Fructose

Q129. The enzyme that hydrolyzes sucrose is: A) Sucrase B) Amylase C) Protease D) Lipase

Q130. Vitamin C is important for: A) Collagen synthesis B) Vision C) Blood clotting D) Calcium absorption

Q131. Which amino acid is aromatic and polar? A) Tyrosine B) Phenylalanine C) Tryptophan D) Valine

Q132. The enzyme class that catalyzes bond cleavage by elimination is: A) Oxidoreductases B) Transferases C) Lyases D) Ligases

Q133. Vitamin B1 deficiency leads to: A) Beriberi B) Pellagra C) Scurvy D) Rickets

Q134. The carbohydrate that is a hexose is: A) Glucose B) Ribose C) Galactose D) Fructose

Q135. The amino acid proline is unique because: A) It is imino B) It is achiral C) It is aromatic D) It is negatively charged

Q136. The enzyme amylase acts on: A) Starch B) Proteins C) Lipids D) Nucleic acids

Q137. Vitamin B2 is also known as: A) Riboflavin B) Niacin C) Thiamine D) Pyridoxine

Q138. The sugar found in DNA nucleotides is: A) Ribose B) Deoxyribose C) Glucose D) Fructose

Q139. The enzyme that hydrolyzes lactose is: A) Lactase B) Amylase C) Protease D) Lipase

Q140. Vitamin D helps in absorption of: A) Calcium B) Iron C) Sodium D) Potassium

Q141. Which amino acid is aromatic and essential? A) Phenylalanine B) Tyrosine C) Glycine D) Serine

Q142. The enzyme class that catalyzes hydrolytic cleavage is: A) Oxidoreductases B) Transferases C) Hydrolases D) Lyases

Q143. Vitamin A deficiency leads to: A) Night blindness B) Pellagra C) Beriberi D) Scurvy

Q144. The carbohydrate that is a disaccharide is: A) Maltose B) Glucose C) Ribose D) Galactose

Q145. The amino acid histidine is classified as: A) Positively charged B) Nonpolar C) Aromatic D) Negatively charged

Q146. The enzyme lipase acts on: A) Lipids B) Proteins C) Carbohydrates D) Nucleic acids

Q147. Vitamin B3 deficiency causes: A) Pellagra B) Beriberi C) Scurvy D) Rickets

Q148. The sugar found in ATP is: A) Ribose B) Deoxyribose C) Glucose D) Fructose

Q149. The enzyme that hydrolyzes proteins is: A) Protease B) Lipase C) Amylase D) Maltase

Q150. Vitamin K is required for: A) Blood clotting B) Vision C) Collagen synthesis D) Calcium absorption

Q151. Which amino acid is classified as nonpolar aromatic? A) Phenylalanine B) Tyrosine C) Tryptophan D) Histidine

Q152. The enzyme class that catalyzes oxidation-reduction reactions is: A) Oxidoreductases B) Transferases C) Hydrolases D) Lyases

Q153. Vitamin B6 deficiency can cause: A) Convulsions B) Night blindness C) Rickets D) Pellagra

Q154. The carbohydrate that is a ketopentose is: A) Ribulose B) Glucose C) Galactose D) Fructose

Q155. Which amino acid is achiral? A) Glycine B) Alanine C) Valine D) Leucine

Q156. The enzyme sucrase hydrolyzes: A) Sucrose B) Maltose C) Lactose D) Cellulose

Q157. Vitamin B2 deficiency symptoms include: A) Cheilosis B) Night blindness C) Rickets D) Scurvy

Q158. The sugar found in FAD is: A) Ribose B) Deoxyribose C) Glucose D) Fructose

Q159. The enzyme that hydrolyzes nucleic acids is: A) Nuclease B) Lipase C) Protease D) Amylase

Q160. Vitamin A is essential for: A) Vision B) Blood clotting C) Collagen synthesis D) Calcium absorption

Q161. Which amino acid is sulfur-containing? A) Methionine B) Glycine C) Alanine D) Valine

Q162. The enzyme class that catalyzes bond formation with ATP hydrolysis is: A) Ligases B) Lyases C) Hydrolases D) Transferases

Q163. Vitamin D deficiency leads to: A) Rickets B) Pellagra C) Beriberi D) Scurvy

Q164. The carbohydrate raffinose is a: A) Trisaccharide B) Disaccharide C) Monosaccharide D) Polysaccharide

Q165. The amino acid threonine is classified as: A) Polar uncharged B) Nonpolar C) Positively charged D) Negatively charged

Q166. The enzyme cellulase acts on: A) Cellulose B) Maltose C) Lactose D) Sucrose

Q167. Vitamin B3 is also known as: A) Niacin B) Riboflavin C) Thiamine D) Pyridoxine

Q168. The sugar found in NAD⁺ is: A) Ribose B) Deoxyribose C) Glucose D) Fructose

Q169. The enzyme that hydrolyzes fats is: A) Lipase B) Amylase C) Protease D) Maltase

Q170. Vitamin K deficiency leads to: A) Poor blood clotting B) Night blindness C) Rickets D) Scurvy

Q171. Which amino acid is aromatic and essential? A) Phenylalanine B) Tyrosine C) Glycine D) Serine

Q172. The enzyme class that catalyzes rearrangement of atoms is: A) Isomerases B) Transferases C) Hydrolases D) Oxidoreductases

Q173. Vitamin E deficiency can cause: A) Fragile red blood cells B) Night blindness C) Pellagra D) Beriberi

Q174. The carbohydrate ribose is a: A) Pentose B) Hexose C) Disaccharide D) Polysaccharide

Q175. The amino acid valine is classified as: A) Nonpolar aliphatic B) Aromatic C) Polar uncharged D) Negatively charged

Q176. The enzyme protease acts on: A) Proteins B) Lipids C) Carbohydrates D) Nucleic acids

Q177. Vitamin B12 is also known as: A) Cobalamin B) Riboflavin C) Niacin D) Thiamine

Q178. The sugar found in RNA nucleotides is: A) Ribose B) Deoxyribose C) Glucose D) Fructose

Q179. The enzyme that hydrolyzes sucrose is: A) Sucrase B) Amylase C) Protease D) Lipase

Q180. Vitamin C is important for: A) Collagen synthesis B) Vision C) Blood clotting D) Calcium absorption

Q181. Which amino acid is aromatic and polar? A) Tyrosine B) Phenylalanine C) Tryptophan D) Valine

Q182. The enzyme class that catalyzes bond cleavage by elimination is: A) Lyases B) Ligases C) Hydrolases D) Transferases

Q183. Vitamin B1 deficiency leads to: A) Beriberi B) Pellagra C) Scurvy D) Rickets

Q184. The carbohydrate glucose is a: A) Hexose B) Pentose C) Disaccharide D) Polysaccharide

Q185. The amino acid proline is unique because: A) It is imino B) It is achiral C) It is aromatic D) It is negatively charged

Q186. The enzyme amylase acts on: A) Starch B) Proteins C) Lipids D) Nucleic acids

Q187. Vitamin B2 is also known as: A) Riboflavin B) Niacin C) Thiamine D) Pyridoxine

Q188. The sugar found in DNA nucleotides is: A) Deoxyribose B) Ribose C) Glucose D) Fructose

Q189. The enzyme that hydrolyzes lactose is: A) Lactase B) Amylase C) Protease D) Lipase

Q190. Vitamin D helps in absorption of: A) Calcium B) Iron C) Sodium D) Potassium

Q191. Which amino acid is aromatic and essential? A) Phenylalanine B) Tyrosine C) Glycine D) Serine

Q192. The enzyme class that catalyzes hydrolytic cleavage is: A) Hydrolases B) Transferases C) Lyases D) Oxidoreductases

Q193. Vitamin A deficiency leads to: A) Night blindness B) Pellagra C) Beriberi D) Scurvy

Q194. The carbohydrate maltose is a: A) Disaccharide B) Monosaccharide C) Polysaccharide D) Pentose

Q195. The amino acid histidine is classified as: A) Positively charged B) Nonpolar C) Aromatic D) Negatively charged

Q196. The enzyme lipase acts on: A) Lipids B) Proteins C) Carbohydrates D) Nucleic acids

Q197. Vitamin B3 deficiency causes: A) Pellagra B) Beriberi C) Scurvy D) Rickets

Q198. The sugar found in ATP is: A) Ribose B) Deoxyribose C) Glucose D) Fructose

Q199. The enzyme that hydrolyzes proteins is: A) Protease B) Lipase C) Amylase D) Maltase

Q200. Vitamin K is required for: A) Blood clotting B) Vision C) Collagen synthesis D) Calcium absorption